

Statistical Mechanics and Field Theory

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The method in (relativistic) QFT, including the second quantization / canonical quantization, path integral, scattering / partition function / correlation function, Feynman diagram, in perturbative methods, representation of particles / symmetry, scale and renormalization group flow ... doesn't depend on the Lorentz invariance;

For example, one can quantize Maxwell eq. without any relativistic consideration, this is easier than QED and was done by physicists in history;

When we apply to some real problems, for example, a solid in our table, it's hard to say the relativistic matters. Thus we can say that the development of QM has two branches in theoretical physics: relativistic and non-relativistic.

QFT

particle physics cosmology
and standard model - AdS-CFT dual

The latter one has statistical nature, ~~and~~ including quantum many-body systems, quantum statistic theory, condensed matter theory. These models are entangled each other and someone may refer all of them as "condensed". (Although in some sense, many models are not so "condensed".)

They can also be viewed as Field Theory, but not relativistic ones, this note discusses the intertwine between two viewpoints: fields are many (statistical) particles.

Rk. Historically, by spin-statistic, physicists expect: use spin- $\frac{1}{2}$ -net (see Ising model next). This idea boost up  日月出矣 旦復旦兮 for QFT. When recently people view things as "all are quantum bit", this indicates

I review a basic fact ~~is~~ ^{seldomly} mentioned in physical texts:
Viewing (quantum) fields are particles, we consider the spinor representation $SO(1,3) \rightarrow \mathbb{H}$ Fock space. Consider the simplest case $SO(3) \rightarrow \mathbb{H}$, the wave functions $|\psi\rangle \in \mathbb{H}$ ~~are~~ do not fill all $\mathbb{H}$, as all ψ are normalized.

Then is $SO(3) \rightarrow SH$? No, we have the physical meaning that, for normalized ψ , $|\psi\rangle \sim |\psi\rangle$, thus it's projective representation $SO(3) \rightarrow PH$, measured by group cohomology as the projective rep are written as $G \xrightarrow{P} PGL(n)$

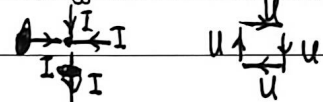
$$g \cdot h \mapsto P(g)P(h) [g, h] \text{ 2-cocycle } \in \mathbb{C}^\times \text{ modulo}$$

The computation of group cohomology does be useful in statistical mechanics, with some model has G ~~is~~ simple.

PART II Statistical models

We start at the classical and easiest one, a circuit diagram

The Kirchhoff first & second laws say that $\sum \vec{I} = 0$ and $\sum \vec{U} = 0$



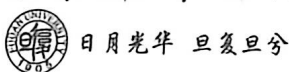
and we have Ohm's principle $U/I = R$

① We have (co)homological interpretations:

$$\begin{aligned} \vec{I} &\xrightarrow{d} d\vec{I} = \vec{V} - \vec{V}' = \vec{V}'' + \vec{V}' \\ \vec{U} &\xrightarrow{\delta} \delta\vec{U} = \sum U_i \end{aligned}$$

gives a duality (Hodge duality)

and the Kirchhoff's first & second law ensures I and U descend to H_0 and H^0 [Frenkel]



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② It can be derived from Maxwell's equations, thus can be viewed as a discrete model of electromagnetism: in circuit diagram, we have $B=0, \rho=0$
 $\Rightarrow \sum I = \oint J = \oint \sigma E = \sigma \int \nabla \cdot E = \sigma \int \rho = 0$

$$\sum U = \oint E = \oint -\frac{\partial}{\partial t} B = 0$$

$$I = \oint J = \sigma \oint E = \sigma U(\sum L) \Rightarrow U/I = \frac{1}{\sigma \sum L} = R$$

Ising model

configuration σ :

We embed such a lattice to different models: each vertex v has $\sigma_v \in \mathbb{Z}_2$

① We can also define d and δ (or $\pm \frac{1}{2}$ spin, here ± 1 for simplicity)

by multiply the spin $\sigma: \sigma_1 \xrightarrow{d} \sigma_2 \xrightarrow{d} \sigma_3 \xrightarrow{d} \sigma_4$

② Viewing all vertex as particles $\sigma_1 \xrightarrow{\delta} \sigma_2 \xrightarrow{\delta} \sigma_3 \xrightarrow{\delta} \sigma_4$

with spin up and down, we study these particles σ_i in statistical method,

the Hamiltonian $E(\sigma) = C_1 \sum_{\langle i,j \rangle} \sigma_i \sigma_j + C_2 \sum_i \sigma_i$

the partition function $Z = \sum_{\sigma} e^{-E(\sigma)}$ and probability $P(\sigma) = \frac{1}{Z} e^{-E(\sigma)}$

③ If the lattice is embedded to 2-dim opt space, then the vertices are finite $(\mathbb{Z}/N\mathbb{Z})^2 \Rightarrow$ all configuration has $2^N \times 2^N$ choices, and we can take $N \rightarrow \infty$ continuous limit

Rk. We may arrange the particles ~~as~~ as it makes the d and delta more natural $\sigma_1 \xrightarrow{d} \sigma_2 \xrightarrow{d} \sigma_3 \xrightarrow{d} \sigma_4$, s.t.

the Hamiltonian is compatible with d and δ ie. $E(\sigma) = C_1 \sum_{\langle i,j \rangle} \delta U + C_2 \sum_i d b$

This is called toric code, I don't know ~~what~~ what's it

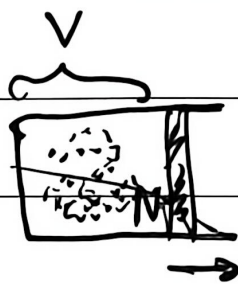
PART III Thermodynamic Limit

When $N \rightarrow \infty$ taking the continuous limit, we can see the "particles" become fully filling the base space.



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Classical thermodynamic limit means that

$N \rightarrow \infty, V \rightarrow \infty$, s.t. $\frac{N}{V}$ is constant

Here we refer it as viewing the continuous limit as number of "particles", when we doing asymptotic expansion

• Large N expansion is thermodynamic limit:

Large N expansion is expanding by the gauge symmetry group $PSL(N)$

this is by the projectized Fock space, $\mathbb{P}^N \leftrightarrow$ Fock space with N

Compare with the ~~any~~ perturbation theory in QFT particles

by coupling constant $\lambda \sim \hbar$, we have $N \sim \frac{1}{\hbar}$

• Conformal block: on the ~~the~~ CFT on a Riemann surface's ~~all~~ ^{some} correlation functions forms a vector space, varying $\mathcal{I}$, it gives a vector bundle ^{chiral} $\mathcal{E}$ on moduli $\mathcal{M}$

Polarize $\mathcal{M} \hookrightarrow \mathbb{P}^N$ after tensor $\mathcal{L}^{\otimes n}$, $n \sim N$

and $H^0(\mathcal{M}, \mathcal{L}^{\otimes n}) \leftrightarrow H^0(\mathcal{M}, \mathcal{E})$ are both called conformal blocks.

In different scenes of mathematics, the conformal blocks are cut off physics, e.g. sheaf $\mathcal{B}_{\text{unif}}(X)$ or D-module $\mathcal{B}_{\text{unif}}(X)$..